

PANDA Component Description: *lcd_simple*

Overview

The *lcd_simple* component implements a simple 2x16 LCD interface. Therefore the 8-bit interface is used. It does not contain any buffers. The component has the ability to generate interrupts if the requested operation is done and the LCD is ready for the next instruction.

Generics

Name	Type	Default	Description
clk_frequency	integer	10'000'000	Clock frequency in Hz

External Ports

Name	Type	Width	Description
rs	std_logic	1	LCD register select signal
rw	std_logic	1	LCD read/write signal
en	std_logic	1	LCD enable signal
dbus	std_logic_vector	8	LCD data bus

Registers

Instruction Register

Name: INST
Address: Base + 0x00

Name	INST
R/W	W
Reset Value	-

The *INST* register is not a real register. If data is written to this register, it is written as instruction to the LCD. A read operation on *INST* returns a random value.

Data Register

Name: DATA
Address: Base + 0x01

Name	DATA
R/W	W
Reset Value	-

The *DATA* register is not a real register. If data is written to this register, it is written as data to the LCD. A read operation on *DATA* returns a random value.

State Register

Name: STATE

Address: Base + 0x02

Name	-	-	-	-	-	-	BUSY	INTON
R/W	-	-	-	-	-	-	R	R/W
Reset Value	0	0	0	0	0	0	0	0

INTON: '1' → An interrupt is generated as soon as the LCD is ready for the next instruction (on every falling edge of *BUSY*)
 '0' → No interrupts are generated

BUSY: '1' → LCD is busy with executing the last instruction
 '0' → LCD is ready for the next instruction

Interrupt Register

Name: INT

Address: Base + 0x03

Name	-	-	-	-	-	-	INT
R/W	-	-	-	-	-	-	R/W
Reset Value	0	0	0	0	0	0	0

INT: '1' → A falling edge of *BUSY* in *STATE* occurred since last clear
 '0' → No falling edge of *BUSY* in *STATE* occurred since last clear

INT can only be cleared. If a '1' is written to it, this does have no effect. *INT* is set when a falling edge of *BUSY* in *STATE* occurs. If *INTON* in *STATE* is set too, an interrupt is generated. *INT* must be cleared by software.

Driver Macros

WAIT_BUSY

Call: *_WAIT_BUSY

Waits until the LCD is ready for the next instruction.